

Yordine Salvador Matuele Chimbutane

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Mechanical engineering undergraduate at PolyU, on the intelligent robotics stream, into working across robotics hardware, embedded control, and fabrication — from CAD and custom PCBs to embedded firmware on competition robots.

EDUCATION

The Hong Kong Polytechnic University — Hong Kong SAR · Sep 2025 – Jun 2029 (expected)

BEng (Hons) Mechanical Engineering — Intelligent Robotics Engineering stream

- **Awards** — Full entry scholarship (2025), Wong Tit-Shing Exchange Scholarship (2026-27).
- **Coursework** — Fundamental AI and Data Analytics; Computer Programming; Applied Engineering Fundamentals; Physics I and II; Calculus, Linear Algebra, Probability and Statistics.
- **Upcoming** — Sensors and Actuators for Mechatronics; Robotics and Computer Integrated Manufacturing; Linear Systems and Controls; Fundamentals of Robotics; Robot Mechanisms and Motion; Advanced Robotics Studio.
- **Activities** — PolyU MantaRay Underwater Robotics Team; Cultural Ambassador; PolyU EAGLE Adventurer 2025/26.

Aga Khan Academy Maputo — Maputo, Mozambique · Aug 2019 – Jun 2025

International Baccalaureate Diploma

- **Score** — **40/45**, top five in cohort. HL Physics, Mathematics, Economics.
- **Awards** — Head of Academy Commendation Award (2024); full merit- and need-based scholarship, MYP and DP (2019–2025).

ROBOTICS EXPERIENCE

PolyU MantaRay Underwater Robotics Team — Hong Kong SAR · Sep 2025 – Present

Person-in-Charge, Buoyancy Float

- Led the float subsystem end to end — watertight, drawn in **Fusion 360**, featuring a custom **PCB**, and with an **ESP32** firmware — taking **3rd place at the MATE ROV 2026 Hong Kong Regionals**.
- Planned the float's depth-hold control firmware with the knowledge of the Archimedes' principle.
- Rebuilt the float for the coming 2027 season: redesigned the hardware with Fusion 360, changes in the electronics including switching the microcontroller to ESP 32 S3 Lolin and integrating it by drawing a new schematic for the internal circuitry.

FIRST Global Challenge 2024 — Team Mozambique · Athens, Greece · Sep 2024 – Jun 2025

Mechanical Team Captain and Primary Driver

- Manufacturing of the national robot within competition specification. Served as mechanical captain while being the youngest member of the delegation — **ranked 40th of roughly 198 competing nations**.
- Drove the robot from the driver station on the **REV Control Hub** platform across tournament matches.
- Ran daily diagnostics and pit repairs so the robot was available for every scheduled round.

PROFESSIONAL EXPERIENCE

Payment Cards Group — Hong Kong SAR · Jul 2026 – Aug 2026

IT Summer Intern

- Built the backend and frontend of a closed-loop compliance pipeline for transaction monitoring.

Belutécnica — Boane, Mozambique · Jul 2024 – Aug 2024

Engineering Intern

- Drafted standardised components and structural layouts in **AutoCAD**, then followed the same parts through cutting, bending, and assembly on the shop floor.
- Practised **MIG** and **TIG** welding and **lathe machining**; joined safety audits and environmental risk reviews.
- Applied **PESTEL** analysis at project initiation and drafted risk assessments to company protocol.

VOLUNTEERING AND LEADERSHIP

PolyU Service-Learning Programme — Ulaanbaatar, Mongolia · Jan 2026 – Jul 2026

Student Engineering Team Member

- Designed off-grid solar systems and DC appliances for Don Bosco Technical Institute, in support of **UN SDG 7 (Affordable and Clean Energy)**.
- Built the project website so the work carries over to the next student cohort.

TECHNICAL SKILLS

- **Robotics and Embedded** — ESP32 and Arduino, embedded firmware, closed-loop control, REV Control Hub, EasyEDA for PCB design
- **CAD and Simulation** — SolidWorks (proficient), Fusion 360 (proficient), AutoCAD
- **Programming** — Python
- **Fabrication** — 3D printing, laser cutting, MIG / TIG welding, lathe machining
- **Languages** — Portuguese (native), English (fluent)